Research proof for review

Optimal fractional vertex cover under vertex arrivals

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In the vertex-arrival model, each new vertex reveals its edges to earlier vertices. A fractional algorithm maintains a value $y_v \in [0,1]$ on each vertex. Values can only increase, and every revealed edge uv must satisfy $y_u + y_v \geq 1$. On an unweighted input, ALG is the final sum of these values and OPT_f is the minimum cost of an offline fractional cover. The competitive ratio is the least R such that $\mathbb{E}[\mathrm{ALG}] \leq R\,\mathrm{OPT}_f$ on every finite input. The expectation is over the algorithm's own randomness.

Theorem. For every $\epsilon > 0$ and every randomized online fractional vertex-cover algorithm, there is a finite, unweighted bipartite vertex-arrival input, independent of the algorithm's realized random bits, with $\mathrm{OPT}_f > 0$ and

$$\mathbb{E}[\mathrm{ALG}] > (B^* - \epsilon)\,\mathrm{OPT}_f.$$

Here $k > 1$ solves $k = \coth k$, $a = (k+1)/2$, $b = (k-1)/2$, and

$$B^* = 1 + a^{a/k} b^{b/k} = 1.9007616968738465\ldots.$$

Together with Wang and Wong's upper bound [1], this gives the optimal competitive ratio on bipartite graphs and on general graphs.

We build the input in two stages. The first stage, called the *core*, starts with a rooted tree and replaces each node by a *pool*: a group of vertices with no edges between them and identical neighbors in the core. For every parent and child in the tree, we put an edge between every vertex of the parent's pool and every vertex of the child's pool. These are all the core edges. For example, a parent with three vertices and a child with two give the six edges of $K_{3,2}$. The second stage adds individual leaves to selected vertices of the core.

The proof keeps track of the worst cost separately according to whether the offline cover includes the root. After a child arrives there are three ways to cover their connecting edge: select the parent only, the child only, or both. A full-history strategy might try to spend less by observing which of the two branches grows first. Sections 2 and 3 show that the adversary can order the two tests so this information has no value at either branch's decisions. Section 4 converts those conditional costs into a feasible function for Wang and Wong's minimax problem. Section 5 then constructs the finite input and completes the theorem. Appendix A gives a separate numerical example.

1 A countable game and its conditional costs

We first describe the core by its tree of pools. A positive rational weight w_P specifies the size of pool P relative to the root, whose weight is one. In any finite core, choose an integer d that clears all weight denominators. Pool P then stands for dw_P ordinary vertices, called twins

because they have the same core neighbors. For example, weights 1 and 2/3 on a parent and child give the $K_{3,2}$ above when $d = 3$. The two pools lie on opposite sides of the bipartition.

At each move the adversary announces the new pool's weight and its parent among earlier pools. The parent may acquire further children at later moves. A public prefix is the ordered list of weights and parent choices seen so far. A causal strategy assigns its response from this prefix alone. The adversary may stop after any move, including the root.

A pool coordinate $y_P \in [0, 1]$ represents the mean of the fractions bought on its twins. Their individual fractions may differ. At the end of a pool's arrivals, summing those fractions gives $dw_P y_P$, and averaging the edge inequalities between adjacent pools P, Q gives $y_P + y_Q \geq 1$. Thus in units of d the total cost is $\sum_P w_P y_P$. In the pool game a response maintains these edge inequalities and may only increase old coordinates. Section 5 obtains such responses from algorithms that see ordinary vertices one at a time.

Fix a rational $0 < \beta < B^* - 1$. We test the proposed competitive ratio $1 + \beta$. An integral pool cover C is a set of pools containing at least one endpoint of each tree edge in the prefix. In the expanded graph it selects every twin in every pool in C. Write $w(C) = \sum_{v \in C} w_v$, where v indexes pools. Define the surplus

$$L(C) = \sum_{v \notin C} w_v y_v - \beta \sum_{v \in C} w_v.$$

If $L(C) > 0$, the algorithm has spent more than $\beta w(C)$ outside the proposed cover. To turn this into a violation of $1 + \beta$, complete the core by presenting one new vertex for each selected twin. This vertex is a private leaf: its only edge joins it to that twin. The $dw(C)$ private edges have disjoint endpoints. Even a fractional offline cover must pay at least one on each of these edges, and the selected twins cover the entire completed graph for exactly $dw(C)$. The online algorithm must pay for the same private edges while retaining its earlier purchases outside C. Hence in ordinary vertex units

$$\mathrm{OPT}_f = dw(C), \qquad \mathrm{ALG} \geq dw(C) + d \sum_{v \notin C} w_v y_v.$$

Lu uses a complete-bipartite extension with the same matching accounting [2]. The edge-arrival construction of Tang and Zhang uses private edges [3].

Start the isolated root at zero. Assume for contradiction that every finite family of core prefixes admits a strategy with $L(C) \leq 0$ for every prefix and cover in the family. Split the covers into two classes according to whether they select the root. We allow a different upper bound on surplus in each class. For real $p \geq 0$, define $F(p)$ as the infimum of the values q for which a **single causal strategy on all core histories** can maintain both promises

$$L(C) \leq p \quad \text{(root outside } C\text{)}, \qquad L(C) \leq q - \beta \quad \text{(root inside } C\text{)}. \tag{1}$$

The strategy space uses countably many coordinates in $[0, 1]$, one vector per public history. Feasibility, monotonicity and the promises at a given prefix are closed finite-coordinate conditions. Compactness, equivalently diagonal subsequence extraction, gives one strategy for all histories. Thus $F(0) \leq \beta$. The root-only inside test gives $F \geq 0$. The infimum is attained by compactness. The set of attainable pairs (p, q) is the epigraph of F. It is closed by compactness of the strategy space and convex by coordinatewise mixing of strategies. Consequently $0 \leq F \leq \beta$, F is nonincreasing and convex, and it is continuous on $[0, \infty)$, including at zero by closedness and monotonicity. No scalar state assumption has been made about the algorithm.

2 Interleaving two finite tests

A *local test* is a pair consisting of a finite rooted input and a terminal cover of that input. Only the sequence of core arrivals is public. The terminal cover is a hidden label. A finite distribution of tests is a tree of public prefixes, with termination and its hidden label as possible final outcomes. Several outcomes may have the same last public prefix. An interleaving merges two such input sequences while preserving the order within each one. Both roots and their connecting edge are already present. A local arrival in the lemma is a subsequent arrival in that component.

Interleaving lemma. Take two distributions, each with finite support and rational probabilities, on tests in disjoint rooted components A, B. Write $\alpha = \Pr(A\text{'s root outside})$ and $\delta = \Pr(B\text{'s root inside})$. If $\alpha \leq \delta$, there is a distribution of interleaved tests with these properties.

1. Both local marginals are preserved, including their entire input and hidden cover. The two selected covers jointly cover the connecting root edge on every outcome of positive probability.

2. At **each local arrival** in either component, conditional on its local public prefix, the joint public past available for that decision gives no further information about the rest of that component's test. This includes its hidden cover.

Construction. In the first local prefix tree mark each node by the conditional probability that its terminal cover excludes its root. In the second tree use the probability that its cover includes its root. Terminal labels are marked zero or one. Each mark is the probability-weighted mean of its children's marks. Implement a multiway transition by a finite binary tree of choices, marking each internal split with its corresponding weighted mean. Only the resulting next public arrival is revealed. Zero-probability outcomes can be dropped. Choose a common denominator M for all marks.

Place two particles at $M\alpha, M\delta$ in $\{0, \ldots, M\}$. They use the **same sequence** of independent fair signs. At each tick add the sign to each interior particle and leave an absorbed particle fixed. For a local binary choice with distinct child marks, wait until its particle first hits either scaled child mark on this lattice and take that branch. From m between child positions $l < u$, the probability of hitting u first is $(m-l)/(u-l)$. This equals the prescribed mass of the upper child since the parent mark is the mean of the two child marks. For equal child marks use a fresh independent coin with the prescribed probability and advance no ticks. Continue this procedure through local arrivals and the final hidden label. Once absorbed, all remaining choices in that local tree have equal marks and use the coin rule.

Merge the local arrivals by tick, keeping local order. Break ties by a fixed component priority. Reveal only actual arriving pools and their parents. Hidden transitions and termination labels are not announcements. The particles remain ordered at every tick, including when one is absorbed. Every required hit occurs almost surely and there are only finitely many local transitions. At termination the two binary marks are ordered. Thus if A's root is outside, B's root is inside. Although the number of private walk ticks is unbounded, the public input ranges over only finitely many local paths and shuffles.

Fix a local public prefix v in component A. Write T_A for its full test and $v \preceq T_A$ for the event that the test reaches v. On this event let H_v be the joint public history up to and

including the arrival at v. For every possible H ending there and every full local test τ, we claim

$$\Pr(T_A = \tau \mid H_v = H) = \Pr(T_A = \tau \mid v \preceq T_A).$$

Fix the other component's independent coins. On reaching v, record the finite word of signs used and the outcomes of the own equal-mark choices made through that arrival. End the record before the next own choice, even if it will use the same tick. For any particular record, the event of reaching v with that record depends only on the signs and own choices it contains. The unused signs and fresh own coins therefore have their original law. The sampler resumes at the node for v, at that node's mark, so the remaining test has precisely the standalone law at v.

Given the other coins, this record also determines H_v. If the record uses m signs, it determines all other arrivals before tick m. Fixed component priority specifies whether the other arrivals at tick m precede this decision. Those arrivals use only the first m signs and the other coins. The conditional law is the same for each such record, so averaging over records and other coins consistent with H proves the claim. The same argument applies on B.

3 Why the budgets must separate

Fix $p \geq 0$ and a strategy attaining (1) at $q = F(p)$. Offer a first child of rational weight $h > 0$. Fix the resulting root coordinates r, z, with $r + z \geq 1$. There are now two rooted components if their connecting edge is set aside. Each can undergo every rational rooted-tree continuation, in any interleaving. Scale the child's subsequent weights by h. Its local surplus will be expressed in units of h. Arrival indices in each local test can be translated canonically to global indices.

We claim there exist **two independent local strategies**, starting at r, z, that meet the global promises against *every compatible pair* of local tests. This is a consequence of the interleaving lemma, even though the original strategy sees both histories.

For each side, a local strategy assigns a response to each of its own prefixes. Independence here means that each assignment uses only that side's history. If there were no such pair, compactness of the product of the two strategy spaces would give a finite list of test pairs whose inequalities no pair of local strategies satisfies. For each test pair the local surpluses L_A, L_B are computed with each component's root normalized to weight one. Thus the required inequality is that $L_A + hL_B$ be at most p or $q - \beta$, according to A's root status. Compatibility here means the two roots are not both outside. The product of local strategy polytopes on these finite prefix trees is compact and convex, and each performance inequality is affine. Write $e_i(x)$ for the excess above the threshold on test pair i, where x ranges over that product polytope. Then $\min_x \max_i e_i(x) > 0$, since the minimum is attained. By finite minimax there are weights $\lambda_i \geq 0$ summing to one for which $\min_x \sum_i \lambda_i e_i(x) > 0$. Approximation within the probability simplex preserves this strict inequality because the finitely many e_i are bounded. We may therefore take rational weights supported on the same compatible test pairs.

Keep only the two local marginals of this distribution. Their ordered root probabilities allow the interleaving construction in Section 2. The expected threshold depends only on A's marginal root status. The expected performance of independent strategies likewise depends only on the two marginals.

To apply separation to the joint strategy, simulate it internally from the fixed coordinates r, z. At each subsequent arrival copy its newborn coordinate y. Replace each older neighbor's current value x by $\max(x, 1-y)$, and keep every other old value. The simulated neighbor value is at least x by induction and at least $1-y$ by feasibility. Thus each coordinate stays at most its simulated value, and all old and new edges are covered. This strategy meets the original promises and changes each component only at its own arrivals.

Let Y_v be its vector on A just after local arrival v. This vector is a function of H_v. Define the local strategy by $\bar{y}_v = \mathbb{E}[Y_v \mid v \preceq T_A]$. The lemma gives

$$\mathbb{E}[Y_v \mid T_A = \tau] = \bar{y}_v \qquad (v \preceq \tau)$$

for every full test of positive mass, including its hidden cover. For a child prefix v' of v, average this equality over the tests reaching v'. On coordinates already present at v this gives

$$\bar{y}_v = \mathbb{E}[Y_v \mid v' \preceq T_A] \;\le\; \mathbb{E}[Y_{v'} \mid v' \preceq T_A] = \bar{y}_{v'}.$$

The local edge inequalities hold by linearity. Start at r and set all coordinates to one on prefixes outside the sampled tree. Do the same on B, starting at z.

Given a full local test, the final local vector equals the vector at its last local arrival, or the fixed initial vector if there was none. Thus the expected surplus conditional on this test equals the surplus of its local mean strategy. Summing over the two marginals, the joint strategy has the same expected excess as this pair of local strategies. Separation makes that excess strictly positive, contradicting the promises on every interleaving.

For the independent strategies let s, t be the suprema of their outside-root surpluses. Pairing each with the other root's root-only inside test gives $s \le p + h\beta$ and $ht \le q$, so both are finite. They are at least r, z by the root-only outside tests. A local strategy starting at a positive coordinate can be simulated from zero by postponing that initial purchase until the first local move. This can only improve its root-only outside test. By the definition of F, the supremal inside surpluses are at least $F(s) - \beta$ on the parent side and $F(t) - \beta$ on the child side. The two inputs can be chosen separately for any compatible pair of root statuses. Taking suprema in the outside/inside and inside/outside inequalities shows that for every $p \ge 0$ and rational $h > 0$ there exist $s, t \ge 0$ satisfying

$$s + h(F(t) - \beta) \le p, \qquad F(s) + ht \le F(p), \qquad s + t \ge 1. \tag{2}$$

The initial promises p need not be reachable from a particular play. Their frontier was defined using the very same rooted input family. At the isolated root we obtain $F(0) \le \beta$.

3.1 A fork that can be checked directly

For a small example, take two adjacent pools of weight one and allow one further child of weight $h = 1/2$ on each endpoint, in either order. Allow a private-leaf completion after any core prefix. The smallest feasible β in this example is $3/5$, so its optimal ratio is $8/5$. Label the endpoints U, V so that V's initial coordinate is at least $1/2$. Attach the first child to U, and let r be U's coordinate then. The cover consisting of U requires $1/2 + h(1-r) \le \beta$. Attach the second child to V and write s for V's coordinate. The two covers that exclude one endpoint require

$$r + h(1 - s) \le \beta(1 + h), \qquad s + h(1 - r) \le \beta(1 + h).$$

At $h = 1/2$, weight the one-child inequality by 3 and the two displayed inequalities by 2 and 1, respectively. The coefficients of r, s cancel, giving $9/2 \le (15/2)\beta$, hence $\beta \ge 3/5$. Conversely keep the isolated root at zero until the connecting edge arrives, then set both endpoints to $1/2$. If a child arrives, raise its parent to $4/5$ and give the child $1/5$. At $\beta = 3/5$ the outside surplus at an endpoint with a child is $4/5 - (3/5)(1/2) = 1/2$. Its inside surplus is at most $(1/2)(1/5) - 3/5 = -1/2$. Without a child the corresponding bounds are $1/2$ and $-3/5$. Every compatible pair therefore has nonpositive total surplus. On completion give each private leaf the complement of its neighbor's current value. Each matched pair now costs exactly one, so the completed cost is at most $(1+\beta)w(C)$. In the blowup by two, a core arrival sees both twins of its parent, whereas a private leaf sees a single vertex. Thus the completion rule can be implemented from the revealed edges alone.

4 Recovering the exact constant

Put $D = \beta - F$. It is nonnegative, concave, and bounded by β. For a differentiability point $0 < p < 1$, convexity and (2) give

$$t \le -F'(p)D(t), \qquad t \ge 1 - p - h\beta.$$

Indeed $ht \le -F'(p)(s - p)$ by the supporting-line inequality, and $s - p \le hD(t)$ with $-F'(p) \ge 0$. For $\eta > 0$ write $g_\eta(v) = v/(D(v) + \eta)$. Concavity and $D(0) \ge 0$ give $D(x) + \eta > (x/y)(D(y) + \eta)$ for $0 \le x < y$, so g_η is strictly increasing. For every sufficiently small rational $h > 0$,

$$-F'(p) \;\ge\; g_\eta(t) \;\ge\; g_\eta(1 - p - h\beta).$$

The right side converges as $h \downarrow 0$, regardless of the choice of t for each h. Hence

$$-F'(p) \;\ge\; \frac{1 - p}{\beta + \eta - F(1 - p)}.$$

Define $f_\eta(u) = \beta + \eta - F(1 - u) > 0$ on the closed unit interval. Integrate from p to 1, using local absolute continuity of convex functions, $F(1) \ge 0$, and continuity to extend to $p = 0$. Then

$$f_\eta(1 - p) + \int_p^1 \frac{1 - u}{f_\eta(u)}\, du \;\le\; \beta + \eta \qquad (0 \le p \le 1). \tag{3}$$

The function f_η is continuous and strictly positive on $[0, 1]$. We show that every continuous function $f : [0, 1] \to (0, \infty)$ satisfying

$$f(1 - p) + \int_p^1 \frac{1 - u}{f(u)}\, du \;\le\; \gamma \qquad (0 \le p \le 1) \tag{4}$$

must satisfy $\gamma \ge B^* - 1$, following Wang and Wong's equalization argument [1, Section 4.1].

First, we equalize the inequality to an exact identity. Set $r_1(p) = f(1 - p)$ and

$$R_1(p) = r_1(p) + \int_{1-p}^1 \frac{1 - x}{r_1(x)}\, dx \;\le\; \gamma.$$

Define two sequences of functions recursively by

$$r_{i+1} \;=\; r_i + \gamma - R_i, \qquad R_{i+1}(p) \;=\; r_{i+1}(p) + \int_{1-p}^{1} \frac{1-x}{r_{i+1}(x)}\,dx.$$

Since $R_1 \le \gamma$, we have $r_1 \le r_2$. If $r_i \le r_{i+1}$ and $R_i \le \gamma$ hold for some i, then

$$R_{i+1}(p) \;\le\; r_{i+1}(p) + \int_{1-p}^{1} \frac{1-x}{r_i(x)}\,dx \;=\; r_{i+1}(p) + R_i(p) - r_i(p) \;=\; \gamma.$$

By induction the sequence $\{r_i\}$ is pointwise nondecreasing and bounded above by γ. Its pointwise limit $r_\infty = \lim_{i\to\infty} r_i$ is continuous and satisfies $R_\infty \equiv \gamma$. By the dominated convergence theorem,

$$r_\infty(p) + \int_{1-p}^{1} \frac{1-x}{r_\infty(x)}\,dx \;=\; \gamma \qquad (0 \le p \le 1).$$

Setting $u = 1-p$, this is $r_\infty(1-u) + \int_u^1 \frac{1-x}{r_\infty(x)}\,dx = \gamma$. Since the integral is differentiable in u, r_∞ is differentiable on $[0,1]$ and satisfies $-r'_\infty(1-u) - \frac{1-u}{r_\infty(u)} = 0$, or equivalently

$$r_\infty(u)\, r'_\infty(1-u) \;=\; u - 1. \tag{5}$$

Next, we solve (5). Replacing u by $1-u$ yields $r_\infty(1-u)\, r'_\infty(u) = -u$. Differentiating the product $r_\infty(u) r_\infty(1-u)$ gives

$$\frac{d}{du}\big[r_\infty(u)\, r_\infty(1-u)\big] \;=\; r'_\infty(u)\, r_\infty(1-u) - r_\infty(u)\, r'_\infty(1-u) \;=\; -u - (u-1) \;=\; 1 - 2u.$$

Integrating from 0 gives $r_\infty(u)\, r_\infty(1-u) = u - u^2 + c$ for the constant $c = r_\infty(0) r_\infty(1) \ge 0$. Write $k = \sqrt{1+4c} \ge 1$, $a = (k+1)/2$, and $b = (k-1)/2$, so $a + b = k$, $a - b = 1$, and $u - u^2 + c = (a-u)(u+b)$. Dividing $r'_\infty(u) r_\infty(1-u) = -u$ by $r_\infty(u) r_\infty(1-u) = (a-u)(u+b)$ gives the separable equation

$$\frac{r'_\infty(u)}{r_\infty(u)} \;=\; \frac{-u}{(a-u)(u+b)} \;=\; -\frac{a}{k(a-u)} + \frac{b}{k(u+b)}.$$

Integrating with the boundary condition $r_\infty(0) r_\infty(1) = ab$ yields the unique solution

$$r_\infty(u) \;=\; (a-u)^{a/k} (u+b)^{b/k}.$$

Evaluating the identity $r_\infty(1-u) + \int_u^1 \frac{1-x}{r_\infty(x)}\,dx = \gamma$ at $u = 1$ gives

$$\gamma \;=\; r_\infty(0) \;=\; a^{a/k} b^{b/k} \;=:\; G(k).$$

Finally, we minimize $G(k)$ over $k \ge 1$. Since $\log G(k) = \frac{a}{k}\log a + \frac{b}{k}\log b$, differentiating with respect to k with $a' = \frac{1}{2}$ and $b' = \frac{1}{2}$ gives

$$\frac{d}{dk}\log G(k) = \frac{k - \frac{1}{2}\log((k+1)/(k-1))}{k^2}.$$

The numerator increases strictly from $-\infty$ (as $k \downarrow 1$) to $+\infty$ (as $k \to \infty$) and vanishes uniquely at $k = \coth k$. The unique minimum of $G(k)$ on $[1,\infty)$ is therefore achieved at this root, giving $G(k) = B^* - 1$ (and strictly beating $G(1) = 1$). Consequently, (3) implies $B^* - 1 \le \gamma \le \beta + \eta$ for every $\eta > 0$. Taking $\eta \downarrow 0$ gives $B^* - 1 \le \beta$, contradicting $\beta < B^* - 1$.

5 Constructing a finite ordinary input

For a requested ϵ, choose a rational $\max(0, B^*-1-\epsilon) < \beta < B^*-1$. The following terminating search specifies finite adversarial families even when a manageable size bound is unavailable.

1. For $n = 1, 2, \ldots$, take all core histories with at most n pools, root weight one, nonroot weights a/b with $1 \le a, b \le n$, and every single-parent choice. Prefixes are shared across all continuations.

2. Test rational linear feasibility with one vector $y \in [0,1]^k$ per public prefix of k pools. Impose monotonicity from its parent prefix, each new edge inequality, zero at the isolated root, and $L(C) \le 0$ for every integral pool cover at that prefix. Exact rational linear programming or Fourier–Motzkin elimination decides each stage. Stop at the first infeasible stage. If every stage were feasible, compactness would give the strategy ruled out by Sections 1–4. The weights used in the scaled child subgames are rational and thus all occur in this exhaustion.

3. Let $d = \mathrm{lcm}(1, \ldots, n)$, the least common multiple of the integers from 1 to n. Every allowed denominator b divides d, so dw_P is a positive integer for every pool P in this stage. For instance, stage $n = 3$ uses $d = 6$ and represents weights 1 and 2/3 by six and four twins. Replace each pool P by dw_P consecutive vertex arrivals, each joined to all vertices of the previously arrived parent pool. Present the root vertices first as isolated vertices. Multiplying costs by the same d preserves the ratio. If the horizon is announced, fix it now at the maximum length of all completed inputs of this stage, and pad shorter ones at the end with isolated vertices. On each fixed expanded prefix take expectations over the algorithm's bits and average over the twins of each pool. Averaging the edge inequalities over all pairs of twins across a parent-child edge gives the pool inequality. Use the same seed on a prefix and its extensions to obtain monotonicity of these means. Set the vector at the root-only prefix to zero and keep the actual means at every later prefix. This preserves feasibility and monotonicity even for algorithms that buy at an isolated root.

4. Infeasibility supplies a prefix and cover C with strictly positive surplus on this evaluation. Finish that prefix with the private matching from Section 1. Both root-only tests have nonpositive surplus, so the prefix contains an edge and $w(C) > 0$. At this prefix the vector equals the algorithm's actual mean. Fix the prefix and its completion before sampling the algorithm's seed. As in Section 1, the private matching and monotonicity outside C give

$$\mathbb{E}[\mathrm{ALG}] \;\ge\; d\left(w(C) + \sum_{v \notin C} w_v y_v\right) \;>\; (1+\beta)dw(C) \;>\; (B^* - \epsilon)\mathrm{OPT}_f.$$

All core and terminal edges are revealed with a new endpoint. The continuum limit in Section 4 tests rational offers of arbitrarily small size under the contrary hypothesis. The output graph itself uses no limiting weights or rounded player responses. The finite search is an existence construction and has no practical near-target runtime guarantee.

The script `scripts/tree_witness.py` implements the construction for a given algorithm. Its function `find_witness(beta, marginals)` evaluates `marginals(core_prefix)` on each scaled family, checks the cover budgets, and returns the completed unweighted arrival list, an offline cover, its optimum and the guaranteed ratio. This permits skipping the LP search for a family hard for every strategy. Given exact comparisons of marginal surpluses and a subcritical target, the search succeeds no later than the first infeasible stage.

A A concrete finite certificate at 1.9005

The certificate describes a particular finite rooted tree-of-pools family without enumerating all its interleavings. The following self-contained recurrence explains how the certificate is generated, why it is a lower certificate, and how it can be checked with exact integer arithmetic. It also checks the direction of the inequalities in (2) without numerical differentiation or evaluating the Wang–Wong function.

A record with rational $a, b \geq 0$ denotes a finite rooted test family on which the promises (1) require $q + bp \geq a$. A root-only record with intercept zero is valid for every nonnegative slope. If a strategy starts the root with a positive coordinate, the same lower bound holds by simulation from zero. At a fork, join two records with lines $a_j - b_j s$ on the parent side and $a_i - b_i t$ on the child side. Offer the child of weight h, then allow all order-preserving shuffles of their core arrivals. A local stop leaves that component idle. Take private completions only after the global core stops, using a compatible pair of terminal covers. Include root-only stops in each component. The finite separation argument of Section 3 applies directly to these two finite test families at any feasible (p, q) because it uses only the global promises on their shuffles. For the resulting local strategies, let s, t be their maximum outside surpluses. The parent record bounds its maximum inside surplus below by $a_j - b_j s - \beta$, and the child record bounds its own by $a_i - b_i t - \beta$. The root-only outside tests give $s + t \geq 1$. Taking maxima for the two mixed root choices gives

$$p \geq s + h(a_i - \beta - b_i t), \qquad q \geq a_j - b_j s + ht.$$

For a desired $b > b_j$ with $bb_i < 1$, choose $h = (b - b_j)/(1 - bb_i) > 0$. Multiply the first inequality by b and add the second. The coefficients of s, t are now both $b - b_j$. Thus the forked family satisfies

$$q + bp \;\geq\; a_j + (b - b_j) - bh(\beta - a_i). \tag{6}$$

Rounding the right side down preserves validity. An alternative operation increases the slope of any old line at unchanged intercept using $p \geq 0$. This uses the same family.

The discrete integer recurrence

To implement (6) without floating-point error, choose a mesh $M = 4000$, a common integer scale $S = 4 \cdot 10^{12}$, and the excess ratio $\beta = 1801/2000$ (corresponding to $1 + \beta = 1.9005$). The root budget in units of S is the integer $B = S\beta = 3{,}602{,}000{,}000{,}000$.

Discretize slopes as $b = \ell/M$ for integers $0 \leq \ell \leq L$, where $L = \lfloor 5M/4 \rfloor = 5000$. Subtree intercepts are represented by integer numerators $A^{(k)}[\ell]$ such that the intercept is $A^{(k)}[\ell]/S$. Candidate child slopes are indexed by $i \in \{0, \ldots, L\}$ on a subgrid with step $\lfloor M/150 \rfloor = 26$.

The recurrence proceeds in sweeps $k = 0, 1, 2, \ldots$:

1. **Base sweep ($k = 0$):** Root-only records have intercept zero: $A^{(0)}[\ell] = 0$ for all $0 \le \ell \le L$.

2. **Sweep update ($k \ge 1$):** Set $A^{(k)}[0] = 0$. For each slope index $\ell = 1, 2, \ldots, L$: The parent slope index is $j = \ell - 1$, with slope $b_j = (\ell - 1)/M$ and intercept numerator $A^{(k)}[\ell-1]$. For each candidate child slope index i from sweep $k-1$ such that $M^2 - \ell i > 0$, setting $h = M/(M^2 - \ell i)$ in (6) yields the integer intercept gain

$$\Delta(\ell, i) \;=\; \left\lfloor \frac{S(M^2 - \ell i) - M\ell(B - A^{(k-1)}[i])}{M(M^2 - \ell i)} \right\rfloor.$$

   Let $\Delta^*(\ell) = \max_i \Delta(\ell, i)$. If $\Delta^*(\ell) > 0$, we choose the maximizing child index $i^*(\ell) = \arg\max_i \Delta(\ell, i)$ and advance the intercept:

$$A^{(k)}[\ell] \;=\; A^{(k)}[\ell - 1] + \Delta^*(\ell).$$

   If $\Delta^*(\ell) \le 0$, no child improves the bound; we set $i^*(\ell) = -1$ (pure slope increase at unchanged intercept) and $A^{(k)}[\ell] = A^{(k)}[\ell - 1]$.

These instructions are completely deterministic and acyclic. They specify relative pool weights h, allowed interleavings, and the finite trees in compressed form. Subtree weights are multiplied by h at each fork and can all be cleared to a common integer denominator on expansion.

At sweep $k = 126$ and slope index $\ell = 4105$ (slope $4105/4000 = 1.02625$), the accumulated intercept numerator reaches

$$A^{(126)}[4105] \;=\; 3{,}602{,}076{,}706{,}847 \;>\; B \;=\; 3{,}602{,}000{,}000{,}000.$$

Since $q + bp \ge A^{(126)}[4105]/S > \beta$, the point $p = 0, q = \beta$ is strictly infeasible on this finite family. The verifier in `scripts/fork_certificate.py` recomputes these integer inequalities from the recorded child choices using Python arbitrary-precision integers. The certificate checks the finite claim conditional on the interleaving lemma. Sections 1–5 prove the limit independently of this example.

From the repository root, run the following command to replay the certificate:

```
python scripts/fork_certificate.py --check experiments/fork-1801-2000.npz
```

Running `python scripts/fork_certificate.py` without arguments regenerates it. The script `scripts/interleaving_check.py` separately checks a finite ordered walk with exact fractions, including coincident arrival ticks and an equal-mark split. A second fixture includes a hidden stop at the root, private splits, and distinct terminal labels with the same root status. It checks both tie priorities. In each case the verifier checks the full local test law at every decision prefix and zero mass on incompatible terminal labels. These finite checks supplement the proof of the general lemma.

References

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